

WHAT PROMPT-SENSITIVITY INDICES ACTUALLY MEASURE: A THREE-AXIS MEASUREMENT FRAMEWORK

Thanos Drossos

Matriculation number 2366736

Karlsruhe Institute of Technology, Karlsruhe, Germany

thanos.drossos@student.kit.edu

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Advisor: Moritz Diener Examiner: Prof. Dr. Gerhard Satzger
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ABSTRACT

Language models answer the same question differently depending on how it is phrased, and the field measures this with a growing set of indices reported in incompatible units, leaving it unclear whether they describe one property or several. We place them on a common ruler taken from functional information, which expresses any such quantity as the negative binary logarithm of a surviving fraction and differs across indices only in the possibility space being counted. On 150 ambiguous questions from AmbigQA, each at two annotator-written specificity levels with ten meaning-preserving paraphrases and ten samples per paraphrase, across three open-weight models, fourteen candidate quantities resolve into three axes: competence, output dispersion, and the share of success variance attributable to phrasing. Two of the three require no new metric, because functional information over the paraphrase space is a monotone transform of accuracy and functional information over the answer space is semantic entropy up to a fixed offset; five published dispersion indices are arithmetic restatements of the latter, agreeing to within 4.4×10^{-16} . The third axis is measured by no existing index and needs the decoding noise subtracted, which a single sample per prompt cannot expose. Two planned interventions separate the axes: disambiguating the question raises competence by 6 to 13 accuracy points and leaves the phrasing share unchanged, while widening the paraphrase generator raises the phrasing share and leaves competence and dispersion flat. Finally, a linear probe on one forward pass flags underspecified questions before any answer is generated and transfers to a differently labelled holdout at an AUROC of 0.670 to 0.678, against matched text baselines of 0.545 to 0.587.

1 INTRODUCTION

Large language models (LLMs) are used through natural language, so the wording of a question is part of the interface. A user asking when a film was released can phrase the question in many equivalent ways, and the choice between them looks arbitrary.

It is not. Model accuracy varies with formatting choices as minor as separators and capitalization (Sclar et al., 2024), with adversarial rewordings (Zhu et al., 2024), and across meaning-preserving paraphrases, where the worst formulation of a question performs substantially below the average one (Cao et al., 2024). Because a single prompt is an unreliable basis for comparing models, benchmarks are increasingly evaluated over prompt sets (Mizrahi et al., 2024; Polo et al., 2024). A second source of variation acts on the question itself: roughly half of real user questions admit more than one valid reading (Min et al., 2020), so a question can fail because the model answered a different reading, not because it lacked the knowledge.

These two sources create a practical problem. Before asking, a user cannot tell whether phrasing or missing detail will decide the outcome, and nothing in the answer marks the difference. A wrong answer looks the same whether the model lacked the knowledge, was unlucky in the phrasing, or answered a different reading of an underspecified question. Suboptimal phrasing therefore costs accuracy silently, and prompt engineering remains trial and error.

The problem persists because the available measurements do not separate these causes. Published indices summarize how much the response distribution moves when the prompt changes, in units that differ from one index to the next (Chatterjee et al., 2024; Errica et al., 2025; Zhuo et al., 2024), which leaves open whether they measure one property of a model or several. Where prompt-induced variance is separated from other variance, it is separated on response embeddings without reference to whether the answer was right (Cox et al., 2025), or aggregated across benchmark items in designs whose scoring is deterministic, so that no sampling-noise term exists to subtract (Romanou et al., 2026). Underspecification of the input has been named as an uncertainty source that existing decompositions do not cover (Kirchhof et al., 2025), and it is rarely manipulated: sensitivity is measured at whatever specificity a benchmark happens to supply. We therefore ask the following research question: *How can sensitivity to phrasing and to semantic change through added context be quantified, per question?*

To compare the indices we need one unit for all of them. We take it from functional information (FI), which measures how rare it is for a configuration to achieve a function, as $-\log_2$ of the fraction that does (Szostak, 2003; Hazen et al., 2007). FI suits prompt sensitivity because the phenomenon is itself a statement about surviving fractions: which share of the ways of asking a question still works, and which share of the answers a question admits the model still entertains. Only the possibility space being counted changes, so quantities reported in different units become commensurable.

Applying that ruler produces the paper’s organizing result. Two of its three instantiations turn out to be quantities the field already reports: FI over the space of paraphrases is a monotone transform of accuracy, and FI over the space of answers is semantic entropy up to a fixed offset. The measurement framework we propose therefore uses the established metrics rather than new ones, and the contribution is the structure: three axes, of which two are already measured under other names, and a third, the share of success variance attributable to phrasing, that no published index isolates. We estimate the model on 150 questions from AmbigQA (Min et al., 2020), each at two annotator-written specificity levels, with ten meaning-preserving paraphrases per level and ten samples per paraphrase, on three open-weight models. Two planned interventions then test the axes against each other: a specificity dial that changes the question, and a generator-width dial that changes the distribution of paraphrases.

The main purpose of this study is to make prompt sensitivity a property that can be reported, validated, and acted upon rather than summarized in one index. This work makes three contributions. First, by placing the published indices on one ruler, we show that five of them are the same measurement and that what remains is three axes. Second, by defining the phrasing share of success variance and estimating it per question with its sampling noise removed, we supply the axis the existing indices do not measure. Third, by predicting underspecification from a single forward pass before any answer is generated, we offer a usable artifact for the practical problem that motivates the study.

2 BACKGROUND AND RELATED RESEARCH

After introducing prompt sensitivity as an empirical phenomenon and the terms this work uses to describe it, this section introduces functional information, the measure from which our metrics are derived. Subsequently, we describe related research on measuring prompt sensitivity and point out the need for this study.

2.1 PROMPT SENSITIVITY OF LARGE LANGUAGE MODELS

A *prompt* is the text supplied to a language model to elicit a response. For a question-answering task, a prompt consists of the question and whatever instructions and evidence accompany it. Because natural language admits many formulations of one question, a single task corresponds to a set of prompts rather than to one. We call two prompts *meaning-preserving paraphrases* of one another when they express the same question, and we call the set of such paraphrases of a question its *paraphrase universe*.

Prompt sensitivity is the dependence of a model’s behavior on which member of that set is used. The phenomenon is robust and has been demonstrated along several dimensions. Accuracy changes with surface formatting such as separators, casing, and the order of options, over ranges large enough to reorder models on a benchmark (Sclar et al., 2024). It changes under adversarial rewordings that preserve meaning (Zhu et al., 2024), and it changes across natural paraphrases, where the worst-performing formulation of a question falls well below the average (Cao et al., 2024). Consequently, benchmark scores computed from one prompt per task carry a component that belongs to the prompt rather than to the model (Mizrahi et al., 2024).

Two properties of the phenomenon matter for its measurement. First, part of the observed variation is not caused by the prompt at all. Generating with a non-zero temperature produces different answers to the same prompt, so a spread measured across prompts contains a decoding component that a single sample cannot distinguish from a phrasing effect. Second, part of the measured sensitivity can be produced by the grading procedure rather than by the model, because a paraphrase may shift the surface form of a correct answer past an exact-match criterion (Hua et al., 2025). Semantic rather than string-based scoring is therefore a precondition for measuring the phenomenon at all.

Prompt sensitivity is distinct from, but interacts with, the *specificity* of the question. An underspecified question admits several valid readings, so a model may answer a reading other than the one intended. About half of the real user questions in Natural Questions (Kwiatkowski et al., 2019) were found to be ambiguous in this sense when annotators were asked to enumerate their readings (Min et al., 2020). The uncertainty that arises because the user did not supply enough information has been named as a source that the standard aleatoric and epistemic decomposition does not cover (Kirchhof et al., 2025).

2.2 FUNCTIONAL INFORMATION

Functional information (FI) counts rarity rather than structure (Szostak, 2003). If only a small fraction of possible configurations performs a function at or above some level, then a configuration that does so is informative in proportion to that rarity; Szostak (2003) illustrates this with RNA aptamers that bind ATP, whose rarity among random sequences corresponds to roughly 37 bits. Hazen et al. (2007) give the general form: for a system of N possible configurations of which $M(E_x)$ attain a degree of function of at least E_x , functional information is $I(E_x) = -\log_2[M(E_x)/N]$ bits. Two features carry over to everything below. FI is a property of an ensemble ranked by function rather than of a single configuration, and it is indexed by the threshold on the function, so it is a curve rather than a number. Wong et al. (2023) extend the construct to any system with a sampleable configuration space and a scalar function measurable on each configuration.

FI is a natural instrument for prompt sensitivity because the phenomenon is already stated in the shape the measure requires. A question corresponds to a set of formulations, a model turns each of them into an answer that is either adequate or not, and the quantity a practitioner cares about is what share of that set works. The same form applies on the output side, where the possibility space is the set of distinct answers a question admits and the function is how sharply one formulation narrows it. Different spaces, one unit.

One property of the measure constrains how its values may be read. Because the denominator counts a space of possibilities, every FI value is relative to how that space is delimited (Elsberry & Shallit, 2011). Corona et al. (2010) therefore formulate FI over a finite, explicitly specified subset of strings rather than an idealized space, which is the form we adopt: our possibility spaces are finite sets that we construct and report, and Section 5.3 returns to what this makes the bit values relative to.

2.3 RELATED RESEARCH ON MEASURING PROMPT SENSITIVITY

Research on prompt sensitivity has produced a range of indices that quantify how strongly the behavior of a model depends on the wording of the prompt. We group the indices by the quantity they summarize.

The largest group reports the *dispersion of outcomes* over a set of prompts that express the same task. Sclar et al. (2024) vary formatting features such as separators, casing, and item order, and report the resulting spread in accuracy. Zhu et al. (2024) construct adversarial perturbations of prompts and report the induced degradation in performance. Zhuo et al. (2024) summarize the disagreement of outputs across prompt variants in a single sensitivity score. Chatterjee et al. (2024) define POSIX from the log-probability that a model assigns to a response generated under one prompt when it is conditioned on another. Errica et al. (2025) define sensitivity as the normalized entropy of the predictive distribution, and consistency as the agreement between the responses to different prompts. These indices are informative, but each is expressed in its own unit, which makes them difficult to relate to one another.

A second group summarizes the *extremes* of the outcome distribution rather than its spread. Cao et al. (2024) evaluate models on the worst-performing prompt of a paraphrase set and show that the gap to average-case performance is substantial for every model they test.

A third group responds at the level of the *evaluation protocol*. Mizrahi et al. (2024) argue that benchmarking on a single prompt is unreliable and call for multi-prompt evaluation. Polo et al. (2024) reduce the cost of such evaluation through item-response models. Hua et al. (2025) caution that a part of the measured sensitivity is an artifact of the grading procedure rather than a property of the model, which motivates semantic rather than exact-match scoring.

Closest to this work is a fourth group that applies *variance decomposition*. The statistical apparatus itself is long established in neighboring evaluation literatures: generalizability theory was proposed for information-retrieval evaluation by Bodoiff & Li (2007) and given its standard treatment there by Urbano et al. (2013), whose variance-component estimator is the one we use, and it entered computational linguistics through the analysis of annotator disagreement (Bayerl & Paul, 2007). For language models, Cox et al. (2025) partition the total variance of response embeddings into an aleatoric component, obtained from repeated sampling at a fixed prompt, and an epistemic component, obtained from variation across paraphrases. They define the prompt sensitivity ratio $\rho_u = U_e/U_t$ as the epistemic share. Romanou et al. (2026) partition the variance of binary correctness into a component attributable to item difficulty and a component attributable to prompt perturbations, and report the perturbation share as a brittleness score per model and benchmark. Their evaluation scores multiple-choice items by output log-probability, so inference is deterministic and the decomposition contains no sampling-noise term. Mustahsan et al. (2025) apply an intraclass correlation to the run-to-run inconsistency of agentic evaluations, with repeated executions rather than paraphrases as the grouping factor. Most recently, full generalizability-theory treatments that estimate a prompt-wording facet separately from a call-to-call replicate facet have appeared for LLM-as-judge pipelines (Messing, 2026) and for corpus-level brand-answer audits (Zatuchin, 2026); both estimate benchmark- or corpus-level summaries rather than per-question quantities, and neither scores against a gold answer. Kunievsky & Evans (2026) formalize the related question-level $\times$ paraphrase-level $\times$ sampling-level split for a continuous extracted value, without a noise-corrected share. Taparia et al. (2026) decompose LLM uncertainty into input ambiguity, knowledge gaps, and decoding randomness — the framing closest to our three axes — but take a single greedy response per paraphrase, so paraphrase-induced variance and generation noise are confounded by construction; the within-cell term that our estimator subtracts does not exist in their design.

A fifth group studies the *underspecification* of the input itself. Kirchhof et al. (2025) name underspecification uncertainty — the user not supplying all information or the exact task — as a research direction that existing aleatoric/epistemic decompositions do not cover. Pecher et al. (2026) report

that a significant portion of observed prompt sensitivity in text classification is attributable to prompt underspecification, at temperature zero and for instruction-side formatting. Hou et al. (2024) decompose uncertainty by generating input clarifications and ensembling over them; the clarifications are consumed inside the estimator, so the ambiguous and the clarified question are never compared as two measured conditions. Keluskar et al. (2024) evaluate ambiguous against disambiguated AmbigQA questions with a single gold answer, which is the closest precedent for our two-level design, without stating how the single gold is selected. On the output side, Cole et al. (2023) show that dispersion over sampled answers tracks question ambiguity on AmbigQA itself, and Staliūnaitė et al. (2026) show that ambiguity information improves downstream error prediction.

Although these studies contribute substantially to the measurement of prompt sensitivity, three gaps remain. First, the reported indices are expressed in mutually incomparable units, so it is unclear whether they measure one property of a model or several; a recent position paper argues informally that mainstream uncertainty indices reduce to one clustering construct (Chen et al., 2026), but no formal reduction has been published. The largest benchmark of uncertainty estimators evaluates every method against a measure of output quality rather than against the other methods (Vashurin et al., 2025), so even there the question of how far the indices duplicate one another is not asked. Second, the decompositions that separate prompt-induced variance from other sources of variance either operate on response embeddings without reference to task success (Cox et al., 2025), aggregate over items and omit the sampling-noise term (Romanou et al., 2026), target the reproducibility of repeated runs rather than the effect of phrasing (Mustahsan et al., 2025), or estimate corpus-level components on outcomes other than task success (Messing, 2026; Žatuchin, 2026). The estimator we use is standard (Urbano et al., 2013); what is not available is its application to gold-referenced task success, per question, with the paraphrase set as the grouping factor. Third, prompt sensitivity is measured at whatever specificity the benchmark happens to supply, because the specificity of the question is not manipulated as an independent variable. This study addresses the three gaps with one information-theoretic ruler, a per-question noise-corrected sensitivity share, and a model-free specificity manipulation.

3 METHODS

We measured the prompt sensitivity of large language models in four steps, as described below. First, we formalized functional information for prompts as the common ruler of the measurement framework. Second, we derived from that ruler a measurement framework consisting of three metric axes and one manipulated variable. Third, we prepared a data set of ambiguous questions with annotated interpretations, and constructed a paraphrase universe for every question. Fourth, we executed a measurement pipeline that samples, scores, clusters, and aggregates model responses into the metrics of the measurement framework.

3.1 A FORMALIZATION OF FUNCTIONAL INFORMATION FOR PROMPTS

To quantify prompt sensitivity, we formalized functional information (FI) as the foundation for all metrics reported in this work. Extant research reports prompt sensitivity through indices in mutually incomparable units, such as accuracy ranges over prompt variants (Sclar et al., 2024), normalized response entropies (Errica et al., 2025), and log-probability divergences (Chatterjee et al., 2024). A common formal basis is needed to state what each index counts and how the indices relate to one another.

FI originates in molecular biology. Szostak (2003) defines the functional information of a molecular activity as the negative logarithm of the probability that a random sequence performs that activity at or above a given level. Hazen et al. (2007) formalize the definition for an arbitrary system of configurations. Let E_x denote a degree of function and let $M(E_x)$ be the number of configurations that attain a degree of function of at least E_x among N possible configurations. Then functional information is

$$I(E_x) = -\log_2 \left[\frac{M(E_x)}{N} \right], \quad (1)$$

measured in bits. Wong et al. (2023) generalize the construct beyond biology and argue that it is well defined for any pair of an enumerable or sampleable configuration space and a scalar function measurable on every configuration. Their broader claim of a natural law is contested in an exchange of letters (Root-Bernstein, 2024; Wong et al., 2024); the dispute concerns evolutionary directionality, not the measure, and does not bear on the use made of Equation 1 here.

Hazen et al. (2007) already work the construct through for *letter sequences*, where the degree of function of a string is the probability that a human receiver understands the message as intended. Our construction replaces that receiver with a language model and the configuration space with a controlled set of meaning-preserving rephrasings. Because an exact $I(E_x)$ would require knowing every possible configuration and its degree of function, Hazen et al. (2007) call their own letter-sequence treatment qualitative; Corona et al. (2010) resolve this by defining $\text{FI}(k) = -\log_2(N_k/|U|)$ over a designated finite subset U with a function measurable on U , and we adopt their formulation. Our U is furthermore *sampled* from a generator rather than designated exhaustively, so the quantity is *generator-relative*: for a proposal distribution G over rephrasings, FI^G is the negative log surviving fraction under G , and values are comparable only within a fixed G . All input-side quantities here carry G = the paraphrase pipeline of Section 3.3. This dependence is a property of any $-\log_2(\text{surviving fraction})$ measure (Elsberry & Shallit, 2011) and we treat it as a limitation (Section 5.3).

We transferred Equation 1 to prompts through three substitutions. A biopolymer sequence becomes a prompt string, the degree of function becomes the correctness of the model response, and the space of possible sequences becomes the set of prompts that express one query. Applying these substitutions yields the general form used throughout this work. For a configuration space $\mathcal{C}$ sampled from G , a function $F : \mathcal{C} \rightarrow [0, 1]$, and a threshold k , functional information is the negative binary logarithm of the fraction of $\mathcal{C}$ that attains function k :

$$\text{FI}^G(\mathcal{C}, F, k) = -\log_2 \frac{|\{c \in \mathcal{C} : F(c) \geq k\}|}{|\mathcal{C}|}. \quad (2)$$

Equation 2 is the ruler of the measurement framework, in the following precise sense. FI_{in} and FI_{spec} are literal instances, counting surviving paraphrases and surviving interpretations respectively. FI_{out} is an instance once the surviving count is taken to be the *effective* answer-space size $2^{H_{\text{sem}}}$, the perplexity of the semantic answer distribution (Section 3.2.3). ρ_F is *not* an instance: it is a variance share *about* the function that the first axis counts (Table 1 lists its unit accordingly), and we do not present it as a bit quantity. Two properties of Equation 2 are relevant for the interpretation of the results. FI is a property of an ensemble ranked by function, not of an individual configuration (Szostak, 2003), and FI is indexed by the threshold k , so it is a curve rather than a scalar (Hazen et al., 2007).

3.2 A MEASUREMENT FRAMEWORK OF PROMPT SENSITIVITY

The measurement framework comprises three metric axes and one manipulated variable. Each axis instantiates Equation 2 on a different possibility space, and the manipulated variable states, in the same unit, how much specificity the question text supplies. Table 1 summarizes the model.

3.2.1 COMPETENCE: THE FI_{in} CURVE

Let q be a query and let U_q be its *paraphrase universe*, the set of prompts that express q without changing its meaning (the construction of U_q is described in Section 3.3). Let $F(x) \in [0, 1]$ be the *degree of function* of a paraphrase $x \in U_q$, defined as the probability that the model answers correctly when the query is phrased as x . With $N_k(q) = |\{x \in U_q : F(x) \geq k\}|$, functional information in prompt space is

$$\text{FI}_{\text{in}}(q, k) = -\log_2 \frac{N_k(q)}{|U_q|}. \quad (3)$$

FI_{in} answers how much phrasing luck a quality level k costs. It is 0 bits when every paraphrase attains k , 1 bit when half of them do, and $\log_2 |U_q|$ bits when a single phrasing does. It is undefined when no paraphrase attains k ; we report such cells separately rather than imputing a value.

Following Hazen et al. (2007), we report FI_{in} as a curve over k rather than as a scalar, and we compute it from the *graded* degree of function (Equation 11), whose eleven attainable values produce a genuine curve; under binary scoring the curve degenerates to a single free number, $-\log_2 \bar{F}$. We evaluated the curve on a uniform grid of 21 thresholds spanning $[0, 1]$. For tabular reporting we also computed the area under the curve, $\text{AUF}_{\text{in}}(q) = \int_0^1 \text{FI}_{\text{in}}(q, k) dk$, by trapezoidal integration, with unreachable thresholds capped at $\log_2(|U_q| + 1)$ before integration so that the area remains finite. Because a substantial share of thresholds is unreachable, the magnitude of AUF_{in} differences scales with this cap convention; we therefore state the cap wherever AUF_{in} is reported, together with the convention-free quantity underneath it, the *censoring rate*: the fraction of thresholds that no paraphrase of the cell attains. As we show in Appendix A.2, AUF_{in} is a deterministic transform of accuracy under binary scoring, which is why we treat the curve, and not its integral, as the representation of this axis.

To capture the shape of the curve in one scalar that binary scoring cannot produce, we additionally computed the *reliability premium*, the two-threshold contrast between usable and perfect reliability:

$$\Delta \text{FI}(q) = \text{FI}_{\text{in}}(q, k_{\text{hi}}) - \text{FI}_{\text{in}}(q, k_{\text{lo}}) = \log_2 \frac{N_{k_{\text{lo}}}(q)}{N_{k_{\text{hi}}}(q)}, \quad (4)$$

with $k_{\text{lo}} = 0.5$ and $k_{\text{hi}} = 1.0$. ΔFI states the additional bits of phrasing rarity that perfect reliability demands over usable reliability. It is identically zero under binary scoring and therefore isolates the information that graded scoring adds.

3.2.2 FORMULATION SENSITIVITY: THE FUNCTIONAL INTRAClass CORRELATION

FI_{in} and its summaries describe how well a model performs. They do not separate two sources of the variation observed across a paraphrase universe: the choice of phrasing, and the stochasticity of decoding. To separate them, we measured the share of the variance in success that is attributable to phrasing, using a one-way random-effects intraclass correlation, ICC(1) (Shrout & Fleiss, 1979; McGraw & Wong, 1996), computed over the $|U_q| \times k$ matrix of binary correctness outcomes with paraphrases as groups. One naming conflict should be stated before the definition, because it is not accidental. Cox et al. (2025) already use the symbol ρ_u for a per-question quantity they call the prompt sensitivity ratio. Our ρ_F is a different quantity computed on a different variable, as we set out below, and we retain the subscript F throughout to mark that it decomposes the variance of the function F .

Let $N = |U_q|$ be the number of paraphrases, let k be the number of responses sampled per paraphrase, and let $\hat{p}_i$ be the observed success rate of paraphrase i over its k samples, with mean $\bar{p}$. The between-group and within-group sums of squares are

$$\begin{aligned} \text{SS}_{\text{between}} &= k \sum_{i=1}^N (\hat{p}_i - \bar{p})^2, & \text{df}_b &= N - 1, \\ \text{SS}_{\text{within}} &= \sum_{i=1}^N k \hat{p}_i (1 - \hat{p}_i), & \text{df}_w &= N(k - 1). \end{aligned} \quad (5)$$

The within-group expression is exact rather than approximate, because the outcomes are binary; the derivation is given in Appendix A.1. With the corresponding mean squares $\text{MS}_b = \text{SS}_{\text{between}}/\text{df}_b$ and $\text{MS}_w = \text{SS}_{\text{within}}/\text{df}_w$, the *functional intraclass correlation* is

$$\rho_F = \frac{\text{MS}_b - \text{MS}_w}{\text{MS}_b + (k - 1) \text{MS}_w}, \quad \rho_F \in [0, 1]. \quad (6)$$

ρ_F states the fraction of the variability in the model’s success that is caused by which phrasing was chosen rather than by decoding noise. A value of 0 means that rephrasing is irrelevant and all observed variability is sampling noise; a value of 1 means that success is fully determined by the phrasing. Following standard practice for variance components, we clamped negative estimates to zero. Where a cell exhibits no variance at all, because every sample of every paraphrase is correct or every one is wrong, Equation 6 is undefined. Such cells are not missing at random: they are exactly the accuracy extremes, so any complete-case analysis of ρ_F is range-restricted on competence. Our primary estimator therefore embeds the same estimand in a hierarchical beta-binomial model, $y_i \sim \text{Binomial}(k, p_i)$ with $p_i \sim \text{Beta}$ of mean μ_c and intra-class correlation ρ_c — for binary outcomes the beta-binomial overdispersion parameter *is* the intraclass correlation — with empirical-Bayes priors on both μ and ρ fitted across all cells by type-II maximum likelihood. The posterior mean of ρ_c is defined for every cell, including the degenerate ones, where it shrinks toward what comparable cells exhibit and carries an honest posterior spread. We report the method-of-moments estimator of Equation 6 alongside, with its coverage, and we report the absolute between-paraphrase variance component $\sigma_B^2 = (\text{MS}_b - \text{MS}_w)/k$, which, unlike the share, is not deflated by a model’s decoding entropy and is therefore the quantity on which models are compared.

ρ_F is the noise-corrected counterpart of the raw performance dispersion that prompt-sensitivity indices commonly report (Zhuo et al., 2024; Sclar et al., 2024). Those indices attribute the full spread across prompt variants to the prompt, whereas Equation 6 removes the component that finite sampling would produce even if every paraphrase were equally good.

Equation 6 is the standard one-way variance-component estimator of generalizability theory, used in retrieval evaluation since Bodoff & Li (2007) and given in this form by Urbano et al. (2013). What differs from the two closest measures in the LLM literature is the variable it decomposes.

Cox et al. (2025) define $\rho_u = U_e/U_t$ as the epistemic share of the total variance of response embeddings, computed without reference to a gold answer. It registers that phrasing changed *what the model said*, whereas ρ_F registers that phrasing changed *whether the model was right*. The two come apart in both directions: a model that expresses the same correct answer in different words scores high on ρ_u and zero on ρ_F , and a model that is identically wrong under every paraphrase scores zero on ρ_u , that is, maximally robust. The estimators also differ. ρ_u is an uncorrected variance share whose expectation under no phrasing effect is $(N-1)/(Nk-1)$ rather than zero, while ρ_F subtracts MS_w and is therefore centered at zero and comparable across sampling budgets. We compute ρ_u alongside ρ_F as a second, embedding-based channel on the same construct.

Romanou et al. (2026) decompose the variance of binary correctness into an item-difficulty and a perturbation component, estimated across items and aggregated per model and benchmark. Their evaluation scores multiple-choice items by output log-probability, so inference is deterministic and the decomposition carries no sampling-noise term. Equation 6 requires exactly that term: MS_w is estimated from the k sampled responses per paraphrase, and it is what separates a phrasing effect from decoding noise. ρ_F is thus the phrasing-attributable share of variance in *task success*, measured within a single question and reported with a posterior uncertainty and a coverage rate. In the vocabulary of global sensitivity analysis, ρ_F is a first-order variance-based sensitivity index of task success with respect to the paraphrase factor, estimated with a noise correction; such Sobol-type indices are established in model attribution (Fel et al., 2021), and we are not aware of a prior application to the paraphrase space of a prompt.

Two statistical properties of Equation 6 on binary outcomes require explicit handling rather than disclosure in passing. First, the Bernoulli variance ceiling ties every variance component to the cell mean, so the *estimate* can co-vary with accuracy even when the true share is identical in every cell. We quantify this with a mechanical null: holding each cell’s mean success rate at its observed value and the true ρ at one constant, re-simulating the full paraphrase-by-sample grid, and re-estimating with Equation 6, the complete-case correlation between $\hat{\rho}_F$ and accuracy that mechanics alone produces spans the observed values in every model and level. Complete-case associations between ρ_F and competence are therefore not interpretable at this design size, which is the second reason the hierarchical estimator is primary. Second, variance-component estimates from few units per facet are unstable (Urbano et al., 2013); a ten-paraphrase universe is far below the sample sizes at which such estimates stabilize, which is why we report per-cell posterior spread, aggregate over questions for all headline claims, and make no per-question point claims from ρ_F alone.

3.2.3 OUTPUT DISPERSION: SEMANTIC ENTROPY AND FI_{out}

The third axis characterizes the responses rather than the prompts. Following Kuhn et al. (2023) and Farquhar et al. (2024), we sampled k responses per prompt, partitioned them into meaning classes by bidirectional natural language inference (NLI) entailment, and computed the *semantic entropy*

$$H_{\text{sem}}(Y \mid X = x) = - \sum_{c \in \mathcal{A}} p_c \log_2 p_c, \quad (7)$$

where p_c is the proportion of the k responses that fall into meaning class c . Semantic entropy is high when the model spreads its answers over many distinct meanings and low when it commits to one. We treat H_{sem} as one instantiation of answer dispersion rather than as the canonical measure: it is a special case of more general semantic-similarity entropies (Nikitin et al., 2024), and any member of that family could serve as the axis representative without changing the structure of the measurement framework.

To express dispersion on the FI ruler, we adopted the entropy-reduction reading of functional information in output space. With $|\mathcal{A}_q|$ the size of the semantic answer space of query q ,

$$\text{FI}_{\text{out}}(x) = \log_2 |\mathcal{A}_q| - H_{\text{sem}}(Y \mid X = x). \quad (8)$$

FI_{out} is the number of bits by which prompt x narrows the answer space relative to a uniform distribution over that space, and it satisfies the identity $\text{FI}_{\text{out}}(x) + H_{\text{sem}}(Y \mid X = x) = \log_2 |\mathcal{A}_q|$. It sits on the ruler of Equation 2 through the perplexity of the answer distribution: $2^{H_{\text{sem}}}$ is the effective number of surviving semantic answers, so $\text{FI}_{\text{out}} = -\log_2 (2^{H_{\text{sem}}} / |\mathcal{A}_q|)$ is the negative log of an *effective* surviving fraction. We rejected the alternative definition $-\log_2 P[\text{correct} \mid x]$, which also has the Szostak form, because its reading inverts: a prompt that rarely produces a correct answer would score as highly informative.

The reference $|\mathcal{A}_q|$ requires care. Estimating it per cell from the number of distinct meaning classes observed makes the yardstick shrink together with the quantity being measured, so that FI_{out} can fall while dispersion also falls. For all comparisons across specificity levels we therefore held the reference fixed at the number of interpretations the data set annotates for the question, m_0 , and report

$$\text{FI}_{\text{out}}^{\text{fixed}}(q) = \log_2 m_0 - \overline{H_{\text{sem}}}(q), \quad (9)$$

where $\overline{H_{\text{sem}}}(q)$ is the mean semantic entropy over the paraphrases of q . Equation 9 is not clamped, because a model that disperses over more meanings than the question admits is informative in itself. Because m_0 is constant within a question, Equation 9 is a within-question affine relabeling of $-\overline{H_{\text{sem}}}$: any paired comparison of $\text{FI}_{\text{out}}^{\text{fixed}}$ across specificity levels is algebraically the same test as the comparison of H_{sem} , and we accordingly report that test once, using $\text{FI}_{\text{out}}^{\text{fixed}}$ only as a unit conversion.

3.2.4 THE MANIPULATED VARIABLE: QUESTION SPECIFICITY

The three axes are outcomes. To move them, we manipulated the specificity of the question itself and measured that manipulation in the same unit. Let m_0 be the number of distinct valid interpretations of the original question and let m_{valid} be the number of interpretations the current wording still admits. *Specificity* is

$$\text{FI}_{\text{spec}} = \log_2 \frac{m_0}{m_{\text{valid}}}. \quad (10)$$

FI_{spec} is the third instance of Equation 2: it counts the fraction of interpretations that survive the wording. It is computed from annotations alone and involves no model, which makes it identical across all evaluated models by construction and therefore admissible as an independent variable rather than as an outcome.

Table 1: The measurement framework. Every metric is an instance of $-\log_2(\text{surviving fraction})$ and differs only in the possibility space it counts.

Component	Metric	Possibility space counted	Unit
Axis 1: competence	$\text{FI}_{\text{in}}(q, k)$ curve	paraphrases of the query	bits
Axis 2: formulation sensitivity	ρ_F	variance of F over paraphrases	share
Axis 3: output dispersion	$H_{\text{sem}}, \text{FI}_{\text{out}}^{\text{fixed}}$	semantic answer space	bits
Manipulated variable	FI_{spec}	interpretations of the question	bits

Note: ΔFI (Equation 4) is a shape summary of axis 1. ρ_F is a variance share rather than a bit count, because it decomposes the function that axis 1 counts.

3.3 DATA GATHERING AND PREPARATION

To manipulate question specificity without changing the task, we used AmbigQA (Min et al., 2020). AmbigQA takes real search queries from Natural Questions (Kwiatkowski et al., 2019) and has human annotators mark the ambiguous ones. For each ambiguous question, the annotators list every distinct interpretation, write a disambiguated question for it, and give that interpretation’s answer. The choice to analyze AmbigQA is supported by three properties. The interpretations, the disambiguated question texts, and the answers are written by annotators rather than generated, so the manipulated variable does not depend on a model; the count of interpretations supplies m_0 directly; and about half of all real queries are annotated as ambiguous, which makes the manipulation representative of practical use. We loaded the `full` configuration of the public validation split, because the test split is not released, and kept only records with at least two annotated interpretations.

3.3.1 SPECIFICITY LEVELS AND GUARDRAILS

For each question we constructed two cells, one per specificity level. Level 0 uses the original ambiguous question, for which $m_{\text{valid}} = m_0$ and $\text{FI}_{\text{spec}} = 0$ bits. Level 1 uses the disambiguated question of one target interpretation, for which $m_{\text{valid}} = 1$ and $\text{FI}_{\text{spec}} = \log_2 m_0$ bits. The target interpretation is selected deterministically from a SHA-256 hash of the question identifier and a fixed seed, so that every model, job, and rerun evaluates the same target.

Two guardrails make the manipulation interpretable. First, *the gold answer is fixed across both levels*: both cells are scored against the target interpretation’s answer, and only the question text changes. Without this constraint, disambiguation would move the scoring target, and any measured gain would be a grading artifact rather than a model effect. Second, *the evidence is identical across levels and across all paraphrases*, so that specificity remains the only manipulated variable.

Target-only scoring has a consequence that must be controlled rather than merely stated: at level 0 the model may answer a different but perfectly valid interpretation and still score zero. A model that answers *some* valid reading with unchanged competence therefore gains $\bar{F}_1 (1 - 1/m_0)$ from disambiguation purely because the grader is told which reading it wanted — a grading lottery, not a model effect. Note that the evaluation protocol of AmbigQA itself scores an ambiguous question against the answers of *all* of its interpretations (Min et al., 2020). We therefore score every cell under *two* gold sets: the pinned target’s answer variants (*target gold*), and the union of every interpretation’s variants (*union gold*). The union set is identical at both levels by construction, so it satisfies the fixed-gold guardrail exactly, and it removes the lottery: its level difference, Δ_{union} , is the specificity effect proper, while $\Delta_{\text{target}} - \Delta_{\text{union}}$ isolates the share of the naive gain that the grading convention manufactures. We report Δ_{union} as the primary effect of the manipulation and the decomposition as a protocol check.

The protocol choice deserves its position in the literature stated plainly. Follow-up work on AmbigQA overwhelmingly preserves all-interpretation scoring, and the common relaxation runs in the *opposite* direction from pinning: crediting a single answer if it matches *any* annotated interpretation is described as common practice (Staliūnaitė et al., 2026; Cole et al., 2023). Evaluating an ambiguous against a disambiguated question with a single gold has one close precedent (Keluskar et al., 2024), which does not state how the single gold is selected when several disambiguations exist; our seeded-hash selection makes that choice explicit and reproducible. The argument that judging re-

sponses to an underspecified query without fixing its context is ill-posed is made independently for open-ended evaluation by Malaviya et al. (2025). Our design takes both sides of this seriously: the pinned target supplies the controlled manipulation, and the union gold supplies the score that the data set’s own protocol licenses.

3.3.2 EVIDENCE

In a closed-book pilot, 84 % of the questions hit a knowledge floor, because the evaluated models of 7 to 8 billion parameters do not know the answers. Under a floor, neither competence nor sensitivity is measurable. We therefore supplied every cell with the annotators’ own search-result snippets shipped with the data set, cleaned of markup and truncated to whole snippets up to a fixed character budget. The snippets are identical across levels and paraphrases, as required by the second guardrail. In addition, we applied a dataset-side filter that keeps only questions whose target answer appears verbatim in the snippet bundle; 52 % of the ambiguous questions pass. The filter uses no model output and therefore introduces no selection on model knowledge. With evidence supplied, answerability rests on reading rather than on recall, which matches the open-book setting in which AmbigQA was constructed.

3.3.3 PARAPHRASE UNIVERSES

The paraphrase universe U_q is not given by the data set; it is constructed, and FI_{in} depends on how it is constructed. For each question and each specificity level separately, we generated candidate rephrasings with a 14-billion-parameter instruction-tuned model, Phi-4 (Abdin et al., 2024), at a sampling temperature of 0.8, using eight writing personas to cover the range of phrasings a user might plausibly produce. The generator is never one of the evaluated models, which avoids a self-generation confound.

Candidates passed two filters. First, an equivalence filter requires *bidirectional* entailment with the original question at a probability of at least 0.9 under a DeBERTa-v3-large NLI model (He et al., 2023) fine-tuned on MultiNLI and related corpora (Williams et al., 2018), so that meaning, and therefore specificity, is held constant within a level. Second, a gold-preservation filter requires that the rephrased question still admits the correct answer, judged by the same generator model at temperature 0. At level 1 the admissible answer is the target interpretation’s; at level 0 it is the union over all interpretations, because a faithful rephrasing of an ambiguous question must preserve every reading. Surviving candidates were deduplicated by Levenshtein distance, and each universe was capped at ten paraphrases. The generated texts are stimuli, not data: no generated text is ever scored as ground truth.

Three properties of this instrument bound what FI_{in}^G can claim. First, meaning-preserving paraphrase is the *mildest* perturbation family in the robustness literature: Romanou et al. (2026) find it produces the smallest accuracy drops of their four families, occasionally improving open-weight models, and warn that LLM paraphrasing may make queries easier by standardizing language. Sensitivity measured through this instrument is therefore a lower bound, and the effects we report are conservative for that reason. Second, the equivalence filter applies an entailment criterion to pairs of *questions*, although entailment is defined on statements with truth values; the filter should be read as a high-precision semantic-match heuristic with an unquantified error rate, backed by the gold-preservation judge, rather than as an application of NLI within its training distribution. Third, we deliberately do not claim that the generator under-samples the phrasing space: the closest published comparison finds LLM paraphrases *more* lexically and syntactically diverse than crowdworker paraphrases, not less. No published prompt-sensitivity metric reports a paraphraser-swap ablation, so generator-relativity is a limitation the field has so far left untested; we therefore include one, together with a direct manipulation of generator width, as a second experimental arm (Section 3.3.5).

3.3.4 MODELS AND SAMPLE

We evaluated three open-weight instruction-tuned models of comparable size from three different families: Llama-3.1-8B-Instruct (Grattafiori et al., 2024), Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), and Qwen2.5-7B-Instruct (Qwen Team et al., 2024). All three were run in process on a GPU cluster rather than through an API, so that decoding parameters, seeds, and hidden states are under

experimental control. Phi-4 was used only as the paraphrase generator and the gold-preservation judge, never as an evaluated model.

From the 2,002 questions of the validation split, the two dataset-side filters of Section 3.3 and Section 3.3.2 retain those with at least two annotated interpretations whose target answer appears in the evidence bundle. We evaluated the first 150 questions of the filtered set in data-set order, which fixes the sample without a random draw and keeps it identical across models and reruns.

The resulting design is a full factorial of question, specificity level, and model. One cell is one triple of question, level, and model, and it contains ten paraphrases with ten sampled responses each, so the main run comprises 900 cells and 90,000 scored responses.

3.3.5 A SECOND MANIPULATION: GENERATOR WIDTH

The specificity manipulation acts on the question and is predicted to move competence. A second manipulation is needed for formulation sensitivity, and the measurement framework says what it must be. Since ρ_F decomposes variance over the paraphrase universe, the quantity that should move it is how far apart the members of that universe are: their *width*, that is, how much the ten rephrasings of one question differ from each other in wording while still meaning the same thing. Width is a property of the generator, not of the question, so we vary it by changing the instructions and the sampling temperature given to the paraphrase generator, holding the question, the evidence, the equivalence filters, and the evaluated models fixed. We measure the realized width as the mean pairwise token edit distance within a universe. To test this prediction, and to test generator-relativity directly, we regenerated paraphrase universes for a 50-question subsample at both specificity levels under four arms. A *narrow* arm uses three minimal-rewording personas at sampling temperature 0.5; the *production* arm is the instrument of Section 3.3.3; a *wide* arm uses eight stylistically diverse personas at temperature 1.0, restricted to fluent rewrites; and a *swap* arm repeats the production instructions and temperature under a generator from a different model family, OLMo-2-13B-Instruct (Team OLMo et al., 2024), which also replaces the gold-preservation judge so that no Phi-4 output enters that arm. All filters, thresholds, and caps are identical across arms; only the proposal distribution changes. Per-arm rejection counts at each filter are persisted, because the filters could in principle compress the manipulation. Five predictions were fixed before any arm data existed: realized width is ordered narrow < production < wide (the manipulation check); σ_B^2 and ρ_F increase with width; accuracy and H_{sem} do not respond; and the swap arm preserves the per-cell structure of ρ_F .

3.4 DATA ANALYSIS

We analyzed each cell in five stages: response sampling, scoring, semantic clustering, metric computation, and aggregation. All stages are seeded, and every model call is cached under a hash of its full request, so that repeated and resumed runs reproduce identical responses.

Sampling. For each paraphrase we drew one greedy response at temperature 0 and $k = 10$ stochastic responses at temperature 1.0, each with its own seed. The greedy response yields a binary score per paraphrase; the stochastic responses yield both the graded degree of function and the input to semantic clustering. Sampling from the same prompt with a fixed seed layout lets a larger- k replication reuse the responses of the smaller one exactly. The choice of $k = 10$ replicates per unit matches the convergence range reported for intraclass correlations on binary LLM outcomes, which stabilize at roughly 8 to 16 trials for structured tasks (Mustahsan et al., 2025); it is adequate rather than generous, and the reliability figures we report reflect that.

Scoring. We scored answers semantically rather than by string matching, because exact match inflates apparent sensitivity with grading artifacts (Hua et al., 2025). An answer counts as correct when a DeBERTa NLI model, given the gold answer as premise and the model answer as hypothesis, assigns an entailment probability of at least 0.7 and a contradiction probability below 0.5. Because AmbigQA supplies several surface variants per answer, the criterion is applied to each variant and combined disjunctively; the same criterion is evaluated under both gold sets of Section 3.3.1, target and union, from the identical cached responses, so the two scorings differ in nothing but the gold set. The graded degree of function of a paraphrase is then

$$F(x) = \frac{1}{k} \sum_{j=1}^k \mathbf{1}[y_{x,j} \text{ is correct}], \quad (11)$$

the fraction of that paraphrase’s k sampled responses that are correct. We recorded a second, more permissive scoring at an entailment threshold of 0.5 as a robustness check. Responses that the model declines to answer are scored as incorrect rather than dropped, so that refusals cannot inflate measured reliability.

Clustering. Semantic clusters were formed over the extracted answer of each response, not the full generation, so that differences in verbosity do not create spurious meaning classes. Two responses join the same class when each entails the other under the NLI model, and the pairwise links are collapsed into equivalence classes. Clustering is performed *once per cell over the pooled responses of all paraphrases*, so that a class identifier denotes the same meaning across paraphrases. Clustering each paraphrase separately would make identifiers incomparable and would inflate $|\mathcal{A}_q|$ and hence FI_{out} . We estimated $|\mathcal{A}_q|$ as the number of distinct classes actually observed. We had initially applied an unseen-class correction (Chao, 1987), but the correction is undefined in the absence of doubleton classes and, with many singleton classes, overestimated the answer space substantially, so we report the directly observed richness, which cannot exceed what was measured. Observed richness at $k = 10$ samples is a downward-biased estimate of the true semantic alphabet, and plug-in entropy inherits that bias; because the bias grows with the true diversity, and diversity is larger for the ambiguous level, the measured H_{sem} drop from level 0 to level 1 understates the true drop, so the cross-level dispersion comparisons are conservative under this limitation. Absolute values of $|\mathcal{A}_q|$ and H_{sem} should nonetheless be read as lower bounds.

Metric computation. From the binary scores we computed the FI_{in} curve on the 21-point threshold grid and its area; from the graded scores we computed ρ_F (Equation 6) and ΔFI (Equation 4); and from the pooled clusters we computed H_{sem} , FI_{out} , $\text{FI}_{\text{out}}^{\text{fixed}}$, and the comparison indices S_τ (Errica et al., 2025), total variation distance consistency, and the variation ratio. POSIX (Chatterjee et al., 2024) requires per-token log probabilities and was computed on a subset, because it costs one teacher-forced pass per ordered pair of prompts. FI_{spec} is attached from the annotations and is not computed from model output.

Aggregation and inference. We report per-cell metrics and aggregate them per model and specificity level. Confidence intervals for $\text{FI}_{\text{in}}(q, k)$ are percentile bootstrap intervals from 1,000 resamples of the paraphrase universe, computed separately at each threshold; intervals for ρ_F are cluster bootstrap intervals from 2,000 resamples of the paraphrases within a cell. To test the effect of specificity, we formed per-question differences between level 1 and level 0 within each model and applied the Wilcoxon signed-rank test; question pairs whose two levels yielded different numbers of paraphrases were excluded, because the capping convention makes their bit values incomparable. Relations among metrics are reported as Spearman rank correlations, computed within a specificity level so that the manipulation is not smuggled into the correlation.

Inference follows a declared family. The primary endpoints of the specificity manipulation are, per model, the change in union-gold accuracy (the effect of interest), the change in target-gold accuracy (the protocol comparison), and the changes in H_{sem} and hierarchical ρ_F — twelve paired Wilcoxon tests in total, for which we report Holm and Benjamini–Hochberg adjusted p -values alongside effect sizes with question-clustered bootstrap confidence intervals (5,000 resamples of questions). $\text{FI}_{\text{out}}^{\text{fixed}}$ is excluded from the family because its paired test is algebraically the H_{sem} test (Section 3.2); counting both would double-count one measurement. The three models are treated as correlated measurements of one question sample, not as independent replications: per-question deltas correlate at Spearman .52–.64 across models, so “significant in all three models” overstates the evidence, and we additionally report the pooled test in which per-question deltas are averaged over models and tested once per endpoint. Reliability of per-cell ρ_F is quantified as disjoint-paraphrase split-half reliability (mean over 200 random five-versus-five splits of the paraphrase universe, Spearman–Brown corrected); the $k=20$ resampling arm is reported only as a sampling-extension check, never as reliability, because the $k=20$ sample contains the $k=10$ sample and a statistic correlated with

its superset measures overlap, not stability. Analysis decisions that materially changed a number, together with the option not taken, are disclosed in the forking-paths appendix.

3.5 PROMPT-LEVEL PREDICTION FROM HIDDEN STATES

The metrics of Section 3.2 require many model calls per question and are therefore diagnostic rather than usable at prompt-writing time. To test whether they are predictable from a single forward pass, we additionally extracted, for every prompt, the hidden state at the last prompt token before any generation, at four fractional depths of the network. Probing hidden states for each of these quantities *individually* is established, and we position every head against its precedent rather than claim the probes as new. Kossen et al. (2024) predict semantic entropy from hidden states with linear probes, at the same last-prompt-token position we use, which anticipates our dispersion head. Zhang et al. (2025) detect question ambiguity from internal activations on AmbigQA itself, with cross-dataset transfer, which anticipates our underspecification head. Predicting from the question alone whether a model will answer correctly is a named task with a public benchmark (Razavi et al., 2025), which frames our reliability head. What none of these connects is the probe signal to the *paraphrase-distribution* quantities of the measurement framework: predicting a question’s formulation sensitivity and its surviving-fraction profile from a single member of its paraphrase universe is, to our knowledge, not previously attempted, and the zero-shot transfer of the underspecification head to annotator-labelled questions under a different labelling mechanism is the evaluation we add. On those features we fitted linear heads that predict whether a question is underspecified, how reliable it is, how dispersed its answers will be, and how phrasing-sensitive it is. All heads were evaluated with grouped cross-validation in which the folds are split by question, so that paraphrases of one question never appear in both training and test data and every reported score is performance on unseen questions. The evaluation is hardened in four ways. Permuted-label controls are full refitted permutation null distributions at every probed layer, not single draws at a reference layer. Layer selection happens inside nested cross-validation, so no reported score benefits from choosing the layer on the test questions. Text-surface baselines, using prompt length, word and character TF-IDF, and the first word, are evaluated under protocols matched to the heads: refit for the in-distribution comparison, frozen for the transfer comparison. Finally, the underspecification head is additionally evaluated zero-shot on an annotator-labelled holdout whose labels were never used for training, with an operating-point analysis at the shipped decision threshold.

4 RESULTS

This section reports the results of the measurement framework on the full factorial design. Section 4.1 tests whether the three axes are statistically distinct. Section 4.2 reports the response of each axis to the specificity manipulation, and Section 4.3 the response to the generator-width manipulation. Section 4.4 reports the measurement properties of ρ_F , and Section 4.5 the prediction of the measured quantities from a single forward pass. Triples of numbers refer to Qwen, Llama, and Mistral in that order, and all inference follows the declared family of Section 3.4.

4.1 THREE DISTINCT AXES

The dispersion indices are one measurement, and this is arithmetic rather than correlation. Evaluated on every cell of the grid, S_τ equals $H_{\text{sem}}/\log_2|\mathcal{A}_q|$ and the variance of FI_{out} equals the variance of H_{sem} , both to a maximum absolute deviation of 4.4×10^{-16} , and $\text{FI}_{\text{out}}^{\text{fixed}}$ equals $\log_2 m_0 - \overline{H_{\text{sem}}}$ exactly; $|\mathcal{A}_q|$, the variation ratio, and total-variation consistency are functionals of the same pooled cluster distribution by construction. Therefore, correlations among these indices are not convergent evidence, and we report H_{sem} as the sole dispersion representative. One boundary case must be disclosed: S_τ is undefined when a cell’s answers form a single cluster, which occurs in 13.8% of cells, so all S_τ analyses condition on at least two clusters.

POSIX is the one comparison index computed from token log-probabilities rather than from the cluster distribution, and it is the one that cannot be assigned: it loads on dispersion and on formulation sensitivity about equally, and the difference is not significant in any model ($p = 0.42/0.63/0.054$, test for dependent correlations; Steiger, 1980).

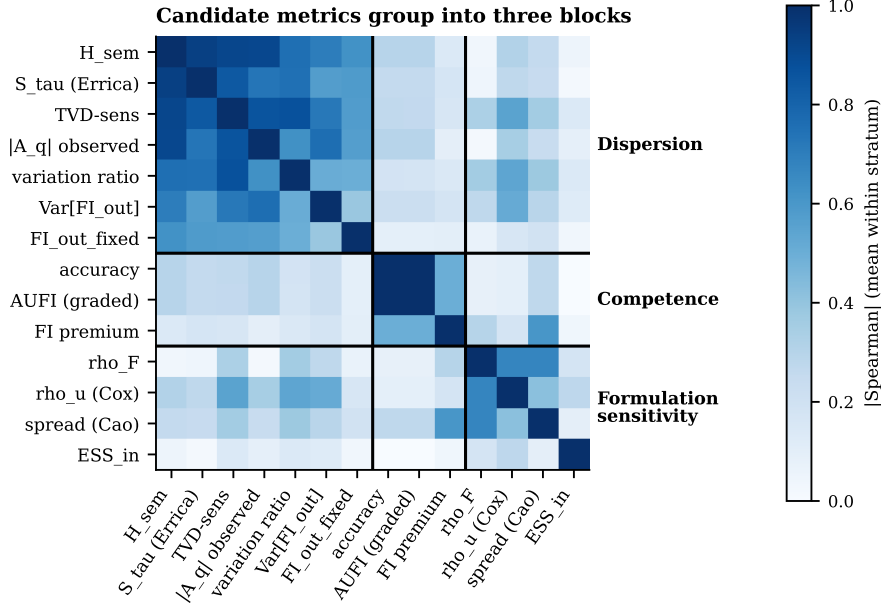


Figure 1: The fourteen candidate metrics, ordered by the axis they were assigned to a priori, on the mean within-stratum Spearman matrix. The dispersion block is near-solid because its members are arithmetic restatements of one clustering. Competence and formulation sensitivity form separate blocks, and every metric loads on the factor of its assigned axis.

Figure 1 shows the resulting structure. The fourteen candidate metrics fall into three blocks, and on the mean within-stratum Spearman matrix the first three principal axes explain 74.6% of the variance, with Horn’s parallel analysis (Horn, 1965) retaining three factors. Because seven of the fourteen inputs are provable functions of one clustering, we treat this as consistent with a three-axis reading rather than as established by it; the evidence that the axes are separate properties comes from the interventions in Sections 4.2 and 4.3, which move them independently.

The axes are distinct, but the data do not support calling them orthogonal. Under the primary estimator, no cross-axis association exceeds $|\rho| = 0.14$ in any model, at either level, under either gold set; however, the sample supports equivalence bounds only of $|\rho| < 0.33$ to 0.47 , so we report bounded non-redundancy rather than independence. One association is consistently non-zero: ρ_F correlates positively with the dispersion factor in all six model-by-level strata under both gold sets (sign test $p = .031$; disattenuated 0.15 to 0.42). In contrast, the complete-case association between ρ_F and accuracy is reproduced in full by a null model in which a constant ρ_F is mechanically coupled to outcome prevalence, with all 24 observed correlations inside the null band, so we treat it as a property of binary intraclass estimation rather than of the constructs.

The stepped shape of per-paraphrase success, the analogue of the “islands of function” reported for molecular sequences, is axis 2 restated rather than a separate finding. The largest adjacent gap in a cell’s sorted success rates exceeds a pure-decoding-noise null in 23.1%/4.0%/15.4% of cells, and this gap statistic tracks ρ_F at Spearman +0.93/ +0.83/ +0.93. Based on these observations, we conclude that *the measurement framework has exactly three distinct axes, the published dispersion indices are a single one of them, and the axes are non-redundant within disclosed bounds rather than orthogonal.*

4.2 RESPONSE TO THE SPECIFICITY MANIPULATION

Table 2 and Figure 2a report the declared endpoint family: the paired per-question change of each axis from the ambiguous to the disambiguated level. Disambiguation raises union-gold accuracy by +0.064/ +0.125/ +0.119; the effect is significant under Benjamini–Hochberg in all three models and Holm-robust in two, and the pooled single-experiment test, which averages each question’s delta

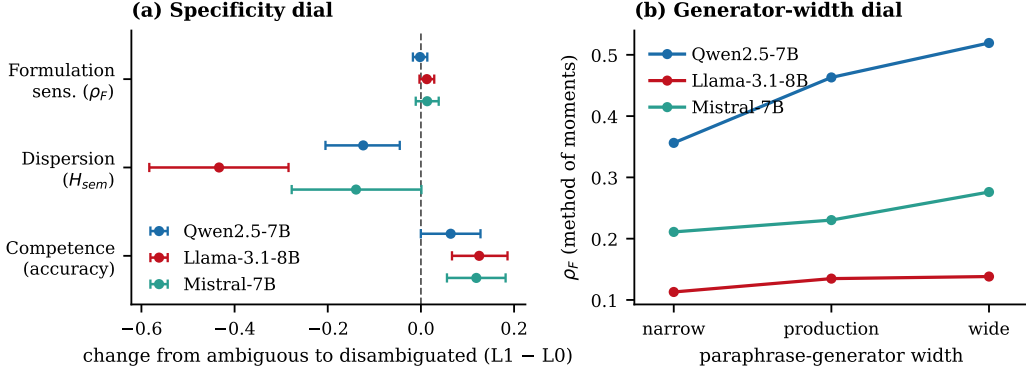


Figure 2: The two interventions move different axes. (a) Disambiguating the question raises competence and lowers dispersion, while formulation sensitivity sits on zero with intervals that exclude changes larger than 0.04. Bars are 95% question-clustered bootstrap intervals on the paired per-question change, $n = 150$ per model. (b) Widening the paraphrase generator raises ρ_F in every model, while competence and dispersion stay flat (Friedman $p \geq 0.47$).

over the three models before testing, gives $+0.103$ ($p = 3.3 \times 10^{-5}$). Under target gold the same manipulation appears two to four times larger, at $+0.225/+0.238/+0.247$; the excess is the grading lottery of Section 3.3.1, and it accounts for 72% / 47% / 52% of the target-gold effect. We report the union-gold numbers as the effect of specificity on competence.

Table 2: Response of the declared endpoints to the specificity manipulation (level 1 minus level 0, paired on $n = 150$ questions per model). Brackets are 95% question-clustered bootstrap confidence intervals. † significant under Benjamini–Hochberg, ‡ additionally Holm-robust, within the family of twelve Wilcoxon tests. Competence responds in every model, H_{sem} responds weakly, and ρ_F does not respond.

Endpoint	Qwen2.5-7B	Llama-3.1-8B	Mistral-7B
Accuracy, union gold (primary)	$+0.064 [-0.000, 0.128]^\dagger$	$+0.125 [0.067, 0.186]^\ddagger$	$+0.119 [0.056, 0.182]^\ddagger$
Accuracy, target gold	$+0.225 [0.161, 0.292]^\ddagger$	$+0.238 [0.183, 0.297]^\ddagger$	$+0.247 [0.183, 0.314]^\ddagger$
H_{sem}	$-0.124 [-0.205, -0.045]^\dagger$	$-0.433 [-0.583, -0.284]^\ddagger$	$-0.139 [-0.277, 0.001]^\dagger$
ρ_F (hierarchical)	$-0.002 [-0.017, 0.014]$	$+0.013 [-0.003, 0.029]$	$+0.013 [-0.011, 0.038]$

Output dispersion responds in the predicted direction in all three models but survives Holm only in Llama. H_{sem} falls by $-0.124/-0.433/-0.139$ bits; the pooled test gives -0.232 bits ($p = 6.1 \times 10^{-5}$). Accordingly, we do not claim a dispersion effect “in all three models”, but a directionally consistent effect that is individually robust in one.

Formulation sensitivity does not respond to the manipulation at all. The change in hierarchical ρ_F is $-0.002/+0.013/+0.013$ ($p = 0.98/0.22/0.48$), on the full $n = 150$ questions per model, and the same null replicates under target gold. This is a properly powered null, not an underpowered one: the confidence intervals exclude changes larger than 0.04 in magnitude in every model. The absolute between-paraphrase variance component σ_B^2 rises with disambiguation under target gold but not under union gold, which identifies that rise as partly a floor artifact of the grading lottery; we therefore interpret σ_B^2 levels, not their level-response.

For axis 1, the convention-free form of the same result is censoring: the fraction of quality thresholds that no paraphrase of a question reaches falls from 55.7%/66.4%/61.9% at the ambiguous level to 32.3%/39.3%/36.2% at the disambiguated level. The corresponding change in the area under the FI_{in} curve is sign-robust under every cap convention but its magnitude ranges from -0.57 to -1.10 bits depending on the cap, so we never report the bit value without its convention (Appendix A.2). Based on these observations, we conclude that *disambiguation raises competence by 6 to 13 points against the full valid-answer set, reduces output dispersion weakly, and leaves formulation sensitivity untouched*.

4.3 RESPONSE TO THE GENERATOR-WIDTH MANIPULATION

The manipulation check passes, with a disclosed compression. Realized width, measured as the mean pairwise token edit distance within a universe, is ordered $4.6 < 8.4 < 9.2$ tokens for the narrow, production, and wide arms, with full per-cell ordering in 66% of cells and narrow below wide at $p = 4.2 \times 10^{-18}$. However, the equivalence filter, not the generator instructions, sets the upper end: the wide arm loses 64% of its candidates to the NLI gate, so the realized gap between production and wide is small, and the dial’s usable range is mostly narrow to production. Any paraphrase-based sensitivity measurement with an equivalence gate inherits this compression.

Table 3: Response of formulation sensitivity to generator width (50 questions, both levels pooled). p is the one-sided Wilcoxon test of narrow below wide. ρ_F is the method-of-moments estimator on covered cells, because the per-arm hierarchical fit is not identifiable on the narrow arm. Both quantities increase with width in every model, while accuracy and H_{sem} stay flat (Friedman $p \geq 0.47$ throughout).

Model	Quantity	Narrow	Production	Wide	p
Qwen2.5-7B	σ_B^2	0.0256	0.0258	0.0301	0.054
	ρ_F	0.356	0.463	0.519	0.004
Llama-3.1-8B	σ_B^2	0.0108	0.0140	0.0153	0.021
	ρ_F	0.113	0.135	0.138	0.079
Mistral-7B	σ_B^2	0.0210	0.0205	0.0249	0.060
	ρ_F	0.211	0.230	0.276	0.051

Table 3 and Figure 2b report the response. The absolute variance component σ_B^2 increases from narrow to wide in all three models ($p = 0.054/0.021/0.060$), and ρ_F increases monotonically across the three arms in all three models, significantly in Qwen ($p = 0.004$). The grading of the effect is itself coherent: the dial moves ρ_F most in the model with the most phrasing sensitivity to amplify (Qwen, $\rho_F \approx 0.5$) and least in the model with the least (Llama, $\rho_F \approx 0.1$). Two estimator notes apply. First, the per-arm hierarchical fit is reported nowhere here, because on the narrow arm, with a median universe of seven paraphrases and a majority of degenerate cells, the mean and correlation hyperparameters are not jointly identifiable, and a simulation confirms this is a regime problem rather than a small-sample bias. Second, matching paraphrase counts across arms by subsampling over five seeds leaves Qwen’s ρ_F increase stable ($p = 0.001$ to 0.013) while the other models’ increases remain directional but not individually significant. In contrast to formulation sensitivity, competence does not respond to width (Friedman $p = 0.68/0.53/0.54$; largest arm-mean gap ≤ 0.02) and neither does dispersion (Friedman $p = 0.47/0.57/0.62$).

The swap arm tests generator-relativity directly. With the paraphrase generator and gold-preservation judge replaced by OLMo-2-13B-Instruct under identical instructions, per-cell ρ_F agrees with the Phi-4 instrument at Spearman $+0.41/+0.27/+0.51$ (all $p \leq .006$), which is at the ceiling set by the instrument’s own split-half reliability, and σ_B^2 agrees at $+0.68/+0.64/+0.80$. The model ranking is preserved under the swapped generator (Qwen $0.36 > \text{Mistral } 0.31 > \text{Llama } 0.13$), while mean levels shift modestly with the slightly narrower accepted universes. Thus, levels of ρ_F are generator-relative, as the formalization of FI_{in}^G requires, but its structure, the ranking of models and of cells, is generator-robust. Together with Section 4.2 this completes a double dissociation: each manipulation moves its own axis and only its own axis. Based on these observations, we conclude that *widening the paraphrase distribution raises phrasing-attributable variance in every model, significantly so in the most phrasing-sensitive one, while leaving competence and dispersion flat, and the structure of ρ_F survives a change of generator family.*

4.4 MEASUREMENT PROPERTIES OF THE SENSITIVITY SHARE

Table 4 collects the measurement properties of ρ_F . Three of them determine how the axis may be used.

First, the estimator choice is not cosmetic. The method-of-moments form is undefined exactly on the all-right and all-wrong cells, so its missingness is outcome-dependent and it covers only half to

Table 4: Measurement properties of ρ_F , by model. The ordering Qwen > Mistral > Llama holds under both gold sets, both estimators, and both paraphrase generators. Reliability is the disjoint-paraphrase split-half correlation; the two predictive checks are evaluated out of sample.

Property	Qwen2.5-7B	Llama-3.1-8B	Mistral-7B
ρ_F , hierarchical (union gold)	0.444	0.091	0.239
ρ_F , hierarchical (target gold)	0.487	0.131	0.252
Variance component σ_B^2	0.0307	0.0143	0.0226
Coverage of the method-of-moments form	45.3 %	66.0 %	57.3 %
Split-half reliability	0.38	0.52	0.57
Agreement between gold sets	+0.53	+0.69	+0.68
Predicts greedy-pass disagreement	+0.51	+0.40	+0.54
Predicts out-of-sample rephrasing payoff	+0.61	+0.39	+0.70

Note: predictive checks are partial correlations controlling for accuracy-extremeness (greedy disagreement) and Spearman correlations on the disjoint second half of the $k = 20$ extension arm (rephrasing payoff).

two thirds of cells. The hierarchical estimator covers every cell with a posterior standard deviation, agrees with the method-of-moments form where both exist (Spearman 0.64 to 0.72), and recovers known values in simulation. We therefore report the hierarchical estimator throughout.

Second, reliability is moderate, which bounds the claims. Per-question point estimates are out of scope; population-level and ranking claims are in scope; and every cross-metric correlation in this paper is attenuated by these reliabilities. Cross-model transfer of per-question ρ_F is weak, at 0.2 to 0.45, so formulation sensitivity is a property of the question–model pair rather than of the question alone.

Third, the construct predicts behaviour outside its own estimation sample. ρ_F estimated from temperature-one samples predicts phrasing disagreement in a separate deterministic greedy pass, and ρ_F estimated from the first ten samples predicts the payoff of rephrasing, the gap between a question’s best paraphrase and its mean, on the disjoint second half of the $k = 20$ arm. Partialling out accuracy leaves the latter essentially unchanged (+0.64/ + 0.38/ + 0.71). Whether rephrasing a question will pay off is therefore decidable, at the population level, before trying. Based on these observations, we conclude that *ρ_F supports population-level and ranking claims with disclosed reliability, and it predicts out-of-sample rephrasing payoff, but per-question point estimates are outside what the instrument licenses.*

4.5 PROMPT-LEVEL PREDICTION FROM A SINGLE FORWARD PASS

In distribution, with grouped cross-validation and nested layer selection, the underspecification head reaches an AUROC of 0.873 to 0.874 in all three models, against a refitted permutation null of 0.50 and a strongest matched text baseline of 0.756 (prompt length). The dispersion head reaches 0.67/0.76/0.77 against text baselines of at most 0.62. The fragility head, the probe for ρ_F itself, does not beat its permutation null in Qwen and Mistral ($p = 0.42$ and 0.50) and reaches 0.63 in Llama ($p = 0.038$), which does not survive any family correction; we report it as a null result, consistent with the reliability ceiling of its target. The reliability-regression head attains rank correlations of 0.35 to 0.51.

Zero-shot transfer is the evaluation that matters for the artifact (Figure 3). Frozen underspecification heads applied to the annotator-labelled holdout, whose labels come from a different mechanism and were never used in training, reach an AUROC of 0.670 to 0.678, while the frozen text baselines reach 0.545 to 0.587.

At the shipped decision threshold, the head flags 67% to 73% of held-out questions at a precision of about 0.66 and a recall of 0.77 to 0.83. With in-domain labels, bag-of-words baselines catch up, so the advantage of the hidden-state head is zero-shot transfer and label efficiency, not raw detection quality. Based on these observations, we conclude that *a single forward pass predicts underspecification and dispersion well before any answer is generated and transfers above matched*

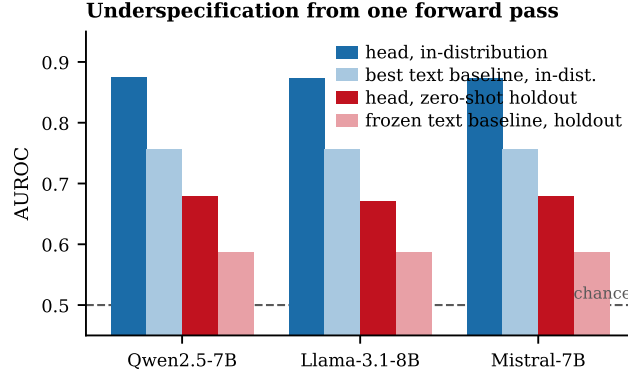


Figure 3: Underspecification predicted from the hidden state at the last prompt token, before any answer is generated. The head beats the strongest matched text baseline in distribution and, frozen, on a holdout whose labels come from a different mechanism. With in-domain labels the bag-of-words baselines catch up, so the advantage is zero-shot transfer and label efficiency rather than raw detection quality.

text baselines without target-domain labels, whereas formulation sensitivity itself is not linearly readable from the prompt’s hidden state.

5 DISCUSSION

This section presents the key findings, elucidates this work’s contributions, and describes its limitations and future research directions.

5.1 PRINCIPAL FINDINGS

Prompt sensitivity is not one measurement. The fourteen candidate quantities we computed occupy exactly three dimensions, and the reduction that establishes this is arithmetic rather than empirical. Normalized response entropy, the variance of FI_{out} , the observed answer-space size, the variation ratio, and total-variation consistency are all functions of one pooled cluster distribution, and they agree with H_{sem} to within 4.4×10^{-16} . A correlation between two of them is therefore not evidence that two constructs agree. This matters for how the field reads its own literature, because convergence among dispersion indices has been available as apparent corroboration while being a restatement of one quantity. The measurement framework answers the research question by naming the three quantities that remain: how well the model does, how much of its success is decided by the phrasing, and how scattered its answers are. Two of the three are quantities the field already reports, and the ruler’s role was to establish that they are the right two and that a third is missing.

The second finding is that each axis responds to its own intervention and to no other. Disambiguating a question raises competence by 6 to 13 accuracy points against the full valid-answer set, reduces output dispersion weakly, and leaves formulation sensitivity untouched within confidence intervals that exclude changes larger than 0.04. Widening the paraphrase distribution reverses the pattern: ρ_F rises in all three models while competence and dispersion stay flat. This double dissociation is the strongest evidence we have that the axes measure separate properties, and it is stronger than a correlation argument, because two quantities can be statistically distinguishable without being separately manipulable. It also settles the interpretation of the one result that first appeared to be a failure. That ρ_F does not move under disambiguation is a properly powered null and a feature of the construct: disambiguation changes what the question asks, not how much its answer depends on the wording.

The third finding concerns what the instrument licenses. The split-half reliability of per-cell ρ_F is 0.38 to 0.57, and cross-model transfer of per-question values is weak, at 0.2 to 0.45. Formulation sensitivity is therefore a property of the question–model pair rather than of the question alone, and

per-question point claims are outside what a ten-paraphrase universe supports. Two out-of-sample checks nonetheless show the construct is not merely internal. Estimated from temperature-one samples, ρ_F predicts phrasing disagreement in a separate greedy pass, and estimated from the first ten samples, it predicts the payoff of rephrasing measured on a disjoint second half, at Spearman $+0.61 / +0.39 / +0.70$. Whether rephrasing a question is worth the attempt is thus decidable at the population level before trying.

Finally, the measured quantities are partly visible in the model’s own representation before it answers. A linear head on the hidden state at the last prompt token separates underspecified from well-specified questions at an AUROC of 0.873 to 0.874 in distribution, and transfers to a holdout whose labels come from a different mechanism at 0.670 to 0.678, against matched text baselines of 0.545 to 0.587. The honest comparison is with Moreno Cencerrado et al. (2025), whose question-only correctness probes reach 0.53 to 0.88 across held-out datasets; our transfer sits inside that band rather than beyond it. The negative result is equally informative: the head for ρ_F itself does not beat its permutation null in two of three models, which is consistent with the reliability ceiling of its target and indicates that formulation sensitivity is not linearly readable from a single prompt representation.

5.2 CONTRIBUTIONS

This work contributes to the measurement of prompt sensitivity in three principal ways.

First, by placing the published indices on one ruler, we enable results reported in different units to be compared, and we show that five of them are the same measurement. Every quantity in the model is $-\log_2$ of a surviving fraction, and the axes differ only in the possibility space counted: the paraphrases of a question, the semantic answers it admits, or the readings its wording leaves open. A useful consequence is that the model does not require new metrics for two of its three axes. Functional information over the paraphrase space is a monotone transform of accuracy, and functional information over the answer space is semantic entropy up to a fixed offset, so competence and dispersion are best reported with the established quantities, and the ruler’s contribution is to show why those two, and only those two, were already sufficient. A position paper has argued informally that mainstream uncertainty indices reduce to one construct (Chen et al., 2026); we supply the reduction. To our knowledge, this is also the first use of functional information to evaluate language models.

Second, by defining formulation sensitivity as the phrasing-attributable share of variance in gold-referenced task success, we supply the axis that the existing indices do not measure. The estimand is what is new: the variance of task success rather than of response embeddings (Cox et al., 2025), per question rather than per benchmark (Romanou et al., 2026; Messing, 2026), and with the within-paraphrase sampling term estimated rather than assumed away. We supply a hierarchical estimator that is defined on all cells, rather than on the 45 to 66 percent where the method-of-moments form exists, and we report the reliability that bounds its use. A mechanical-coupling null then shows that the apparent association between ρ_F and accuracy is a property of binary intraclass estimation rather than of the constructs. The generator swap is, to our knowledge, the first paraphraser-swap ablation reported for a prompt-sensitivity metric, and it shows that levels of ρ_F are generator-relative while its structure is generator-robust.

Third, by predicting underspecification from one forward pass, we offer an artifact for the problem that motivates the study. A user who cannot tell whether a question is underspecified receives a warning before the model answers, at no generation cost and without target-domain labels. This turns the measurement framework from a description of models into a check that can sit in front of one.

5.3 LIMITATIONS AND FUTURE RESEARCH

Every bit value we report is relative to a reference class. FI_{in} counts the fraction of a paraphrase universe that succeeds, and that universe is produced by one generator under one set of instructions and filtered by one equivalence gate. The measure is therefore defined with respect to that proposal distribution, not to the space of formulations a population of users would produce. This is the reference-class dependence that has been raised against functional information generally (Elsberry & Shallit, 2011), and it is why we adopt the finite-subset formulation (Corona et al., 2010) and report

the universe rather than treat it as given. The generator swap bounds the problem without removing it: rankings survived, levels shifted. Survey methodology names the assumption our equivalence gate asserts, that two wordings differ in form only, and shows that it can fail when a rewording moves the referent rather than the framing (Huber & Paris, 2013). No published prompt-sensitivity metric reports a paraphraser-swap ablation, so this is a field-wide gap rather than a local one, and the natural next step is a rule-based or human-written generator arm. More ambitiously, the reference class is a design parameter rather than a nuisance: a deployment could choose the generator that models its own user population and read the resulting bits as costs for those users.

The design measures sensitivity to meaning-preserving paraphrase of factoid questions, and this bounds the construct in three ways. Paraphrase is the mildest of the perturbation families surveyed by Romanou et al. (2026), so our estimates are a lower bound on sensitivity to prompt variation in general; format perturbations and typographic noise are not represented. The equivalence gate compresses the width dial by rejecting 64 percent of the wide arm’s candidates, and it rejects longer rewritings more often, so the usable range of that dial runs from narrow to production rather than to genuinely wide. Any paraphrase-based measurement with an equivalence gate inherits this compression.

Statistical scope is limited by the size of the design rather than by the estimator. Variance components estimated from ten paraphrases sit in the regime that Urbano et al. (2013) identify as unstable, where roughly fifty units per facet are needed for stable estimates, which is why we report population-level and ranking claims and not per-question point estimates. Intraclass estimates on binary outcomes are additionally coupled to outcome prevalence (Gulliford et al., 2005); our null model reproduces all twenty-four observed complete-case associations between ρ_F and accuracy, so those associations are uninformative about the constructs in either direction at this design size. Relatedly, $|\mathcal{A}_q|$ and H_{sem} at ten samples are lower bounds on the true semantic richness, which makes our cross-level comparisons conservative but leaves absolute values bounded rather than estimated. With 150 questions, three models of 7 to 8 billion parameters, and one dataset, we make no claim about frontier-scale models, and the zero-shot probe holdout is our only evidence of external validity.

Three threats to validity are conceded rather than resolved. Consistency-based uncertainty estimators have been argued to be limited under genuine ambiguity (Tomov et al., 2025). Our design escapes the sharpest form of that critique, because it decomposes the variance of task success and takes the interpretation counts from annotators rather than from a model. That escape is an argument, however, not a measurement. A paraphrase that breaks the answer extractor scores identically to a reasoning failure (Mehta, 2026); we score unparseable responses as incorrect, which concedes the confound. Finally, the grading protocol itself carries a choice: crediting only the target interpretation makes disambiguation appear two to four times more effective than crediting any valid reading, and we report the union-gold numbers as primary for exactly that reason.

Beyond closing these gaps, three directions follow. A specificity ladder with more than two rungs would identify a dose-response curve in bits, which a two-point design cannot. Applying the model to non-factoid and multi-step tasks would test whether three axes remain sufficient where the answer space is not a short span. Most broadly, the artifact suggests a role for measurement frameworks as an audit layer in front of deployed systems, where a prompt is checked before it is answered and a rephrasing agent consults ρ_F to decide whether rewording is worth the call.

6 CONCLUSION

The wording of a question changes whether a language model answers it correctly, and a user choosing between equivalent formulations has no signal telling them that the choice matters. The field has responded with a growing collection of sensitivity indices reported in incompatible units, which makes it difficult to tell whether they describe one property or several.

This work treated the problem as one of measurement. Functional information supplies a single unit, $-\log_2$ of a surviving fraction, in which the competing quantities can be expressed and compared; only the possibility space being counted changes. In that unit the candidate metrics resolve into three axes: how well a model does, how much of its success is decided by the phrasing, and how scattered its answers are. Two of the three need no new metric, because functional information over the

paraphrase space is accuracy in a logarithmic wrapper and functional information over the answer space is semantic entropy offset by a constant. Five published dispersion indices are arithmetic restatements of the latter. The axis that no existing index measured is the phrasing share of success variance, which requires estimating and subtracting the decoding noise that a single sample per prompt cannot expose.

The two interventions support reading the three axes as distinct properties rather than as three views of one. Disambiguating a question raised competence by 6 to 13 accuracy points, reduced answer dispersion weakly, and left formulation sensitivity unchanged. Widening the distribution of paraphrases raised formulation sensitivity while leaving competence and dispersion flat. Each dial moved its own axis and no other. The measurements are relative to the paraphrase distribution that produced them and are reliable enough for population-level and ranking claims rather than per-question verdicts, and we report both bounds rather than leaving them implicit.

For practice, the most useful consequence is that some of this can be seen before the model answers. A linear head on a single forward pass separates underspecified from well-specified questions and transfers to questions labelled by a different mechanism, which turns a diagnostic property of models into a check that can sit in front of one. Prompt sensitivity is measurable, it is three things rather than one, and knowing which of the three is at fault is what makes it actionable.

AI USE STATEMENT

Generative AI was used in this work, and this section documents that use, as required by the ICLR 2027 AI Policy for Authors and by the KIT Guidelines for the Use of Generative AI. Under the disclosure categories of the ICLR policy, AI assistants were used to implement methods, that is, to write and refactor parts of the experimental code base; to provide feedback on research methodology; to summarize and help identify related literature; to draft and edit parts of this paper for readability, grammar, and phrasing; and to format references.

The research contribution was not delegated. The research question, the design of the measurement framework and of the two interventions, the choice of data set and guardrails, and the interpretation of the results are the author’s own. Literature was assessed by the author against the primary sources: every citation in this paper was checked field by field against the publisher record or the original PDF, and several bibliographic errors and one incorrect novelty claim introduced by AI-assisted drafting were corrected in that process. All generated code was reviewed, tested, and executed by the author, and every number reported here comes from the committed experimental artifacts rather than from a model’s description of them.

Consistent with KIT’s guidelines, AI output was treated as a draft requiring verification rather than as a result. The author takes full responsibility for the content of this paper, including any part that was drafted or suggested with AI assistance.

REPRODUCIBILITY STATEMENT

The code, configuration, and analysis scripts for this work are available at

<https://github.com/ThanosDrossos/PromptSensitivityFI>

The repository contains the paraphrase-generation and evaluation pipeline described in Section 3.3, the metric implementations of Section 3.2, the analysis scripts that produce every table and figure in this paper, and the per-cell result files from which they are computed. Model identifiers and decoding parameters are given in Section 3.3.4, and the selection of the target interpretation is a seeded hash of the question identifier (Section 3.3.1), so the sample and the levels reproduce exactly. All model calls are cached under a hash of the full request, and every stage is seeded, so a resumed or repeated run returns identical responses. The two derivations that the main text states without proof are given in Appendix A.2 and Appendix A.1.

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A DERIVATIONS

A.1 THE WITHIN-GROUP SUM OF SQUARES FOR BINARY OUTCOMES

Equation 5 states the within-group sum of squares of the functional intraclass correlation as $\sum_i k \hat{p}_i(1 - \hat{p}_i)$. This expression is exact, not an approximation, because the outcomes are binary.

Let $y_{ij} \in \{0, 1\}$ be the correctness of the j -th of k responses sampled from paraphrase i , and let $\hat{p}_i = \frac{1}{k} \sum_{j=1}^k y_{ij}$ be the observed success rate of that paraphrase. Because $y_{ij}^2 = y_{ij}$ for binary outcomes,

$$\begin{aligned} \sum_{j=1}^k (y_{ij} - \hat{p}_i)^2 &= \sum_{j=1}^k y_{ij}^2 - 2\hat{p}_i \sum_{j=1}^k y_{ij} + k\hat{p}_i^2 \\ &= k\hat{p}_i - 2k\hat{p}_i^2 + k\hat{p}_i^2 \\ &= k\hat{p}_i(1 - \hat{p}_i). \end{aligned} \tag{12}$$

Summing Equation 12 over the N paraphrases gives SS_{within} as stated. The practical consequence is that ρ_F can be computed from the per-paraphrase success rates alone; the individual outcomes y_{ij} need not be retained.

A.2 THE AREA UNDER THE FI_{in} CURVE UNDER BINARY SCORING

Section 3.2.1 states that $AUFI_{\text{in}}$ is a deterministic transform of accuracy when the degree of function is binary. The derivation explains why this work reports the curve rather than its integral.

Let $F(x) \in \{0, 1\}$ for every $x \in U_q$, let $N = |U_q|$, and let $a = N_1/N$ be the fraction of paraphrases that answer correctly, that is, the accuracy of the paraphrase universe. For any threshold $k \in (0, 1]$, the set $\{x : F(x) \geq k\}$ equals $\{x : F(x) = 1\}$, so $FI_{\text{in}}(q, k) = -\log_2 a$ is constant on that interval. At $k = 0$ every paraphrase qualifies and $FI_{\text{in}}(q, 0) = 0$.

Writing $c = -\log_2 a$ and integrating by the trapezoidal rule over the uniform grid $k_j = j/20$ for $j = 0, \dots, 20$ with spacing $h = 0.05$ gives

$$AUFI_{\text{in}} = h \left[\frac{FI_{\text{in}}(q, k_0) + FI_{\text{in}}(q, k_{20})}{2} + \sum_{j=1}^{19} FI_{\text{in}}(q, k_j) \right] = 0.05 \left[\frac{0 + c}{2} + 19c \right] = 0.975 c. \tag{13}$$

$AUFI_{\text{in}} = 0.975 \cdot (-\log_2 a)$ is therefore a strictly monotone function of accuracy, and the rank correlation between the two is -1 by construction. The graded track softens the identity but does not remove it. $AUFI_{\text{in}}$ consequently carries no information about sensitivity that accuracy does not already carry, which is why the measurement framework reports the FI_{in} curve as the representation of the ability axis and treats its integral as an alias of accuracy.

A.3 RELATION OF FI_{out} TO NORMALIZED RESPONSE ENTROPY

For a task with C mutually exclusive answer classes and no semantic clustering, the sensitivity measure of Errica et al. (2025) reduces to the response entropy normalized by its maximum, $S_\tau(x) = H(Y | X = x) / \log_2 C$. Substituting $|\mathcal{A}_q| = C$ into Equation 8 gives

$$FI_{\text{out}}(x) = \log_2 C - H(Y | X = x) = \log_2 C (1 - S_\tau(x)). \tag{14}$$

On closed-class tasks, FI_{out} is thus an affine rescaling of $1 - S_\tau$, expressed in bits rather than as a unit-free ratio. The two quantities separate only on open-ended generation, where the answer classes are not given and must be recovered by semantic clustering.

B ESTIMATOR CONVENTIONS

Table 5 records the conventions applied when a metric is undefined or unbounded. Each convention is chosen to be conservative with respect to the hypotheses under test.

Table 5: Conventions for undefined and unbounded metric values.

Quantity	Degenerate case	Convention
$FI_{in}(q, k)$	no paraphrase attains k	undefined; capped at $\log_2(N + 1)$ before integration
$AUFI_{in}$	unreachable thresholds	capped values integrated
ΔFI	no paraphrase attains k_{lo}	0 bits (the curve has no shape there)
ρ_F	no variance in the cell	missing, not zero; coverage reported
ρ_F	negative variance estimate	clamped to 0
FI_{out}	$ \mathcal{A}_q $ underestimated	clamped to 0
FI_{out}^{fixed}	dispersion exceeds $\log_2 m_0$	not clamped; negative values reported
FI_{spec}	$m_0 = m_{valid} = 1$	0 bits

Note: N denotes the number of paraphrases in the cell and m_0 the number of annotated interpretations of the question.

C ADDITIONAL MATERIAL